Testing the Sattelzeit as a Network Event

A method note on `koselleck-networks`

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Contents

1 Context

The historian Reinhart Koselleck argued that the period 1770 – 1830 (the **Sattelzeit**, German for "saddle period") was not simply an era in which many words happened to change meaning at the same time. He claimed it was a moment when the whole vocabulary of politics and society reorganized **as a system** – concepts shifted together, in relation to one another, rather than drifting independently one at a time. This has always been argued in prose, from close reading of texts. It has never been tested computationally.

Ryan Heuser's *Computing Koselleck* took the first step towards such a test: he trained a separate word-embedding model for each historical period and measured how far each individual word moves from one period to the next. He found that this movement is unusually large around 1770 – 1830, which supports the idea that something real happened in that window. But his method looks at one word at a time. It cannot say whether words moved **together**, as a system reorganizing around a shared structure, or whether each word's shift is its own independent event that simply happens to cluster in time.

This project adds that second, missing test. Instead of asking how far a single word moves, it asks whether the **groupings** among words – the way the whole vocabulary is organized into clusters of related meaning – change unusually around the same decades. If Koselleck is right, we should see the cluster structure itself reorganize at the Sattelzeit, not just individual words drifting off on their own.

2 Method

The method reuses the same per-period training that Heuser's approach uses, so the two studies remain directly comparable, and adds a network layer on top that a word-by-word approach cannot produce.

1. **Split the corpus into time windows.** The corpus is divided into even 20-year windows, from 1500 to 1900. The windows are uniform across the whole timeline, not just around the Sattelzeit: if the method measured more finely right where reorganization was expected and more coarsely everywhere else, it would be biased toward finding exactly what it was looking for. Heuser's own results, in fact, show a second spike of word-level change around 1905 – 1925, unrelated to the Sattelzeit – a pattern that a Sattelzeit-only sampling scheme could never have picked up.

2. **Train a language model per period.** For each time window, a model (**word2vec**) is trained independently, learning a numeric representation of every word from how it is actually used in that period's text: words that appear in similar contexts end up

with similar representations. Training a separate model per period, rather than one model updated continuously, is what makes it possible to later measure both individual word drift and whole-system reorganization: a word's meaning only makes sense relative to the other words trained alongside it, in that same period.

3. **Build a word-similarity network.** From those representations, a network is built for each period: every word is a node, linked to its 15 closest neighbours by **cosine similarity** (a standard measure of how alike two words' representations are). A fixed number of nearest neighbours is used instead of a similarity threshold, because at this vocabulary scale a fixed threshold produces networks that are almost fully connected – tens of millions of edges per period, unusable and with no meaningful structure to cluster. Purely grammatical words (articles, prepositions, pronouns) and single-letter tokens are excluded as network nodes: the interest is in words that carry conceptual content, not the grammatical scaffolding of the language (those grammatical words are still used during training itself, since they help place every other word correctly).

4. **Detect communities.** A **community detection** algorithm (**Leiden**) is run on each network, grouping words that are densely connected to one another. In practice, these groups function as semantic fields: sets of words that, in that period, are used in similar ways to each other more than to the rest of the vocabulary.

5. **Repeat detection at several levels of detail.** Community detection does not have one single "correct" answer: a parameter called **resolution** controls how many groups appear and how large they are. For that reason every network is processed at seven different resolution settings, and any claim about reorganization has to hold up across most of those settings, not just one chosen after the fact. This is a hard rule of the project: no finding is reported if it only holds at a single, convenient resolution.

6. **Compare consecutive periods.** For each pair of consecutive periods, the comparison is restricted to words that appear in both (each period has its own vocabulary). The best possible match between one period's groups and the next period's groups is found first – group numbers do not mean the same thing from one period to the next, so the correspondence has to be established before anything can be compared – and then the fraction of shared words that still ended up in a different group, despite that best match, is measured. That fraction, how much of the shared vocabulary "migrated" to a new group, is what this project calls the **migration fraction** – its central measurement of how much reorganization happened between two periods.

7. **Test whether reorganization concentrates at the Sattelzeit.** The hypothesis predicts that the migration fraction should be unusually high specifically for the transitions that fall inside or close out the 1770 – 1830 window, compared to every other transition across the full 1500 – 1900 timeline.

8. **Dictionary cross-check (not yet implemented).** As a second line of evidence, independent of the network entirely, the plan is to take the words that change group right at the pivot and check whether the OED or other period dictionaries independently record a dated sense change at the same moment. This layer is not yet designed: the corpus mixes British and American sources, and matching each source against the wrong regional dictionary would invalidate the comparison, so the choice of dictionary per source has to be resolved first.

2.1 Corpus

The primary source is the **Text Creation Partnership (TCP)**: curated, manually corrected transcriptions with no OCR noise, covering 1500 – 1800 (EEBO-TCP, ECCO-TCP, Evans-TCP). Since the Sattelzeit extends to 1830 and TCP stops at 1800, the plan is to supplement 1800 – 1900 with Project Gutenberg. HistWords (pre-trained historical embeddings) is used only as an independent sanity check, not as primary evidence.

3 The interactive explorer

Alongside the pipeline, a small web tool makes the per-period networks directly explorable, without reading raw numbers. It has two views of the same underlying data.

Graph explorer. A period slider drives a force-directed network diagram: nodes are words, their size reflects how connected they are, and their color marks which community they belong to in that period. A search box focuses the view on one word and its neighbourhood; moving the slider one period at a time shows that neighbourhood evolving, and a change in a word's color between periods marks a real change of community, not just a coincidence of the diagram's layout. A side panel explains the method in plain terms and shows, for the currently selected resolution, how the migration measurement holds up across all seven resolution settings – the same robustness check described above, made visible rather than just claimed.

Word search. The same underlying data, presented as a plain table instead of a diagram: type a word and a period, and get back its closest neighbours by similarity, each neighbour's community, and whether the searched word's own community changed since the previous period. Useful when the actual numbers matter more than the picture.

Every community shown in either view carries a short, plain-English label (for example "Government & Law", "Moral Character & Virtue"). These labels were generated by having a language model read each community's most representative words and summarize the shared theme in a few words – a convenience for a human reader, not a finding in themselves. Roughly half of all communities were honestly flagged, by the same process, as **“mixed”**: the corpus is multilingual (English, Latin, Law French, Welsh, Scots, and more) and early modern, and a community detected by this method sometimes groups words by shared grammar – verb conjugations, comparative forms, spelling variants – rather than by shared topic. Both views surface this caveat directly next to the labels, rather than presenting every label as equally reliable.

4 Current limitations

Two limitations shape what can be claimed right now, and both are visible in the tool itself rather than hidden.

Corpus coverage. The corpus does not yet cover 1800 – 1900 with the same density as earlier periods, since the Gutenberg supplement described above has not been sourced yet. The periods with the least surviving text are, not coincidentally, the ones closing the Sattelzeit window, so any comparison across that boundary currently rests on a thinner sample than the rest of the timeline.

Label reliability. The community labels described above are a single, unvalidated read-through by a language model, not a checked ground truth – useful for orientation, not for citation.